Adjoint Shape Optimization of a Conformal 2.4 GHz Patch Antenna: Solve Counts Against Tuned Gradient-Free Search

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Abstract

We design the copper of a probe-fed 2.4 GHz patch antenna on a 1.6 mm FR-4 board bent to a 100 mm radius with geode-fem, a finite-element solver whose discrete shape adjoint gives the gradient of a band objective at one factorization per frequency, and count solves to fixed worst-of-band $|S_{11}|$ levels on the coarse optimization model against tuned gradient-free search. With 38 shape parameters the adjoint reaches −9.5 dB in 35 solves; the fastest of the separable CMA-ES, full CMA-ES and NEWUOA runs to reach that level needs 385 (11×), and the fastest to reach −9 dB needs 7.7× the adjoint's solves (full CMA-ES, one seed). Against the fastest run to reach each level, the ratio is 1.1× at −9 dB and 2.4× at −9.5 dB with 4 parameters, and 3.3× and 2.9× with 14. L-BFGS with a finite-difference gradient follows the adjoint's iterates at 39 times its solves, behind CMA-ES, so the savings come from the adjoint. Re-centred on the rim-resolved model, the design covers 2.400–2.4835 GHz at −10 dB on the three most refined models computed, including −10.46 dB on Palace at third order and −10.32 dB on Palace at second order on a rim-refined mesh, which leaves 1.9 MHz of lower-edge margin. A larger air margin raises the lower edge by 3.0–4.3 MHz, so we estimate that the design misses the lower band edge by a few megahertz once refined further. Its radiation efficiency is 0.61–0.63 with perfectly conducting copper.

1 Introduction

Gradient-based inverse design is routine in nanophotonics, where differentiable solvers such as CEVICHE (Hughes et al., 2019), the Meep adjoint solver (Hammond et al., 2022), SPINS-B (Su et al., 2019) and FDTDX (Mahlau et al., 2026) optimize a permittivity density on a structured grid. Shape derivatives of scattering parameters for finite-element models with ports are older (Georgieva et al., 2002; Nikolova et al., 2004a; Webb and Akel, 2000) and are available in commercial tools (Ansys, Inc., 2025; COMSOL AB, 2023). A gradient is worth its implementation cost only if it buys something that gradient-free search does not. With two nearly independent parameters, finite differences or a scaling law are cheap. With many coupled parameters, finite differences cost one solve per parameter and gradient-free search slows with dimension (Rios and Sahinidis, 2013), which is where an adjoint should pay.

This paper measures that on one antenna. A 2.4 GHz probe-fed patch is bent around a cylinder of radius $R_b = 100$ mm. The board is a uniform 1.6 mm FR-4 shell that is not designed; up to 38 smooth parameters move the copper outline, the corners and the feed. geode-fem, a differentiable finite-element solver with a discrete shape adjoint, optimizes the band objective by L-BFGS. On the same problem, objective, start and volume guard we run separable and full-covariance CMA-ES (Hansen and Ostermeier, 2001; Ros and Hansen, 2008) and NEWUOA (Powell, 2006; Zhang et al., 2023) over a range of settings, the same L-BFGS with a finite-difference gradient, and, for two

optimizers, the independently developed solver Palace (Palace developers, AWS Center for Quantum Computing, 2023) as the forward model, and count solves to fixed objective levels.

Three results follow.

- **The adjoint cuts the solve count, and the cut grows with the number of parameters** (Section 5): from close to even with 4 parameters, where three gradient-free runs end lower than the adjoint, to 11× at −9.5 dB and 7.7× at −9 dB with 38, each against the fastest gradient-free run to reach that level. A finite-difference gradient on the same L-BFGS path costs 39 times the adjoint's solves, so the savings are the adjoint's, not the gradient's as such; given enough solves, gradient-free search reaches the same level.
- **The forward model's error sits at the patch rim** (Section 4). Palace on geode-fem's own meshes agrees with geode-fem at first and second order, and at third order with geode-fem's finest rim-refined mesh, and a port-free eigenmode places the matched dip 16 MHz above the cavity resonance.
- **A design re-centred on the refined model covers the ISM band at −10 dB on the three most refined models computed** (Section 7), with a probe of realizable width (1.6 mm, modelled as a lumped port). Its lower-edge margin shrinks under further refinement and a larger air margin, and we estimate that it misses 2.400 GHz by a few megahertz; we do not claim that it meets the ISM band.

Evidence tiers. We keep three tiers apart. A *bounded* result is a proven two-sided bound; no result in this paper is of that kind. A *model-dependent* result is a value computed for a stated discrete model (mesh, element order, absorber and port model). An *estimated* result is an extrapolation to the mesh limit, given with its band and never called a converged value.

2 Related Work

Finite-element S-parameter sensitivities with ports. Design sensitivities of the scattering matrix for tetrahedral edge elements were derived by Webb and Akel (2000), extended to the frequency response by Webb (2002) and used in gradient optimization of microwave devices by Nair and Webb (2003). Georgieva et al. (2002) give adjoint S-parameter sensitivities that need only the solver's field solutions, so that commercial solvers can supply them; adjoint and self-adjoint formulations are reviewed by Nikolova et al. (2004a) and Nikolova et al. (2006b), and continuum shape sensitivity for high-frequency devices is given by Choi et al. (2011). HFSS computes S-parameter derivatives with respect to geometric parameters by reusing the LU factors (Ansys, Inc., 2025; Vardapetyan et al., 2008), and COMSOL documents adjoint shape optimization of waveguide devices on a deformed mesh (COMSOL AB, 2023; Jensen, 2022). For antennas, Ghassemi et al. (2013) evolve a microstrip antenna with an FEM adjoint sensitivity and a small set of control vertices. Toivanen et al. (2009) obtain antenna shape sensitivities from a method-of-moments solver by automatic differentiation, and Toivanen et al. (2010) optimize ultra-wideband antennas whose outlines are parametrized by splines, a close prior combination to our spline outline and gradient descent. Jacobsson and Rylander (2010) apply gradient-based shape optimization to conformal array antennas.

Gradient-free and surrogate-assisted antenna optimization. Evolutionary search is the common gradient-free route in antenna design, and surrogate models cut its cost: Liu et al. (2014) combine differential evolution with Gaussian-process surrogates, Zhou et al. (2021) use trust-region

parallel Bayesian optimization, and Akinsolu et al. (2020) review machine-learning-assisted methods. The closest precedent for our measurement is aerodynamic: Zingg et al. (2008) compare a genetic algorithm with an adjoint method on aerodynamic shape problems and find that the genetic algorithm needs 5 to 200 times as many function evaluations. We did not run a surrogate-assisted or Bayesian optimizer.

Variable-fidelity antenna optimization and space mapping. Space mapping aligns a cheap coarse model with an accurate fine model and optimizes the coarse one (Bandler et al., 1994, 2004; Koziel et al., 2006). In antenna design the coarse model is often the same electromagnetic solver on a coarse discretization, corrected by frequency scaling or by response-shape correction (Koziel, 2010; Koziel and Ogurtsov, 2014), and adjoint sensitivities have been used inside trust-region space mapping (Koziel et al., 2013). Our correction is the simplest member of this family: a frequency shift fitted between two discretizations of the same solver.

Automated shape differentiation and structured grids. Shape derivatives of finite-element models can be generated automatically in UFL and Firedrake (Dokken et al., 2020; Ham et al., 2019; Paganini and Wechsung, 2021) and in NGSolve (Gangl et al., 2021). Wang and Anderson (2011) give a discrete Maxwell shape adjoint on unstructured meshes in two dimensions, in the time domain and without ports. On structured grids, conductors are designed by density-based topology optimization (Erentok and Sigmund, 2011; Hassan et al., 2014a,b) and with FDTD shape sensitivities (Nikolova et al., 2004b, 2006a). A plain Yee grid staircases a curved conductor (Cangellaris and Wright, 1991; Häggblad and Runborg, 2014), which locally conformal FDTD avoids (Benkler et al., 2006; Dey and Mittra, 1997; Mittra and Yu, 2000; Zagorodnov et al., 2003). Subpixel-smoothed projection gives the Meep density adjoint subpixel-accurate boundaries (Hammond et al., 2025; Romano et al., 2026), though its material-grid adjoint excludes metals (Meep developers, 2026); Tidy3D offers autograd shape gradients and conformal PEC averaging (Flexcompute, Inc., 2026); shape parameters can be mapped to a fixed grid (Hooten et al., 2025; Liu and Bonar, 2024); and boundary-integral methods carry recent PEC shape-sensitivity work (Arens et al., 2026; Balouchev et al., 2024; Takahashi, 2025).

Conformal and cylindrical patch antennas. The cylindrical-rectangular patch has cavity-model analyses (Krowne, 1983; Luk et al., 1989), and nonplanar microstrip antennas and conformal arrays are treated in the monographs of Wong (1999) and Josefsson and Persson (2006). We use a full-wave model and do not compare with these analyses; the curvature here is mild (Section 4).

Position of this work. What this paper adds is a measurement on an open-radiator problem with a moving feed: the solve cost of a finite-element shape adjoint against tuned gradient-free optimizers and a finite-difference gradient at 4, 14 and 38 parameters. We do not claim a first EM shape adjoint, and we make no performance comparison with structured-grid methods.

3 Method

3.1 The open-radiator forward model

The forward problem is the driven frequency-domain curl-curl equation, discretized with Nédélec edge elements of order $p = 1$ or $p = 2$ on an unstructured tetrahedral mesh. Lengths are in millimetres and the solver works with $\omega = k_0 = 2\pi f/c$ in rad/mm. The assembled system is the

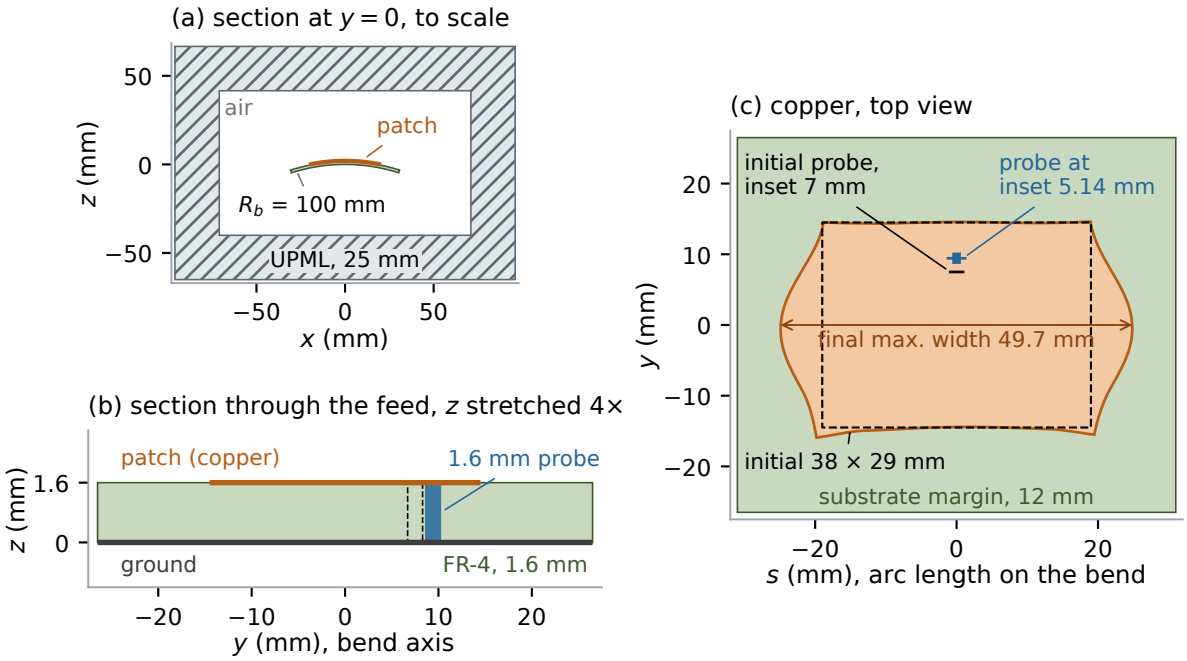


Figure 1: The fixture and the final copper design. (a) Section at $y = 0$, to scale: the FR-4 board (green) on the $R_b = 100\,\text{mm}$ cylinder with the patch on top, inside the air box and the 25 mm box UPML (hatched). (b) Section through the feed at $s = 0$, vertical axis stretched four times: patch, 1.6 mm FR-4, ground, and the 1.6 mm probe, final (filled) and initial (dashed). (c) Top view in the chart (s, y): the initial $38 \times 29\,\text{mm}$ patch (dashed), the final copper (filled) and the probe positions (bars). The board is the same in both. Drawn from `baseline_2g4_L1.toml` and `recentre/geometry/rbG_L1_copper_38_sm57_w16.json`.

complex-symmetric pencil

$$A(\omega)\,\mathbf{x} = \left[\, K(\nu) - \omega^2 M(\varepsilon) + \tfrac{j\omega}{Z_s}\, S_p \,\right] \mathbf{x} = \mathbf{b}, \tag{1}$$

with curl-curl stiffness K, mass matrix M and lumped-port surface term S_p. The radiating boundary is a box UPML of thickness $t = 25\,\text{mm}$: per-element tensor weights Λ enter K (as $\nu = \Lambda^{-1}$) and M (as $\varepsilon = \varepsilon_r \Lambda$), with the stretch $\varsigma = 1 - j\sigma_0 (d/t)^2/\omega$ per axis at depth d and $\sigma_0 = 1$. The FR-4 has $\varepsilon_r = 4.4$ and $\tan\delta = 0.02$, and the conductors are perfect electric conductors (PEC). The port is a rectangle of width w_p along y and length $l_p = 1.6\,\text{mm}$ across the substrate, with sheet impedance $Z_s = R_0\, w_p / l_p$ and $R_0 = 50\,\Omega$. The reflection coefficient follows from the port voltage V:

$$S_{11} = \frac{Z - R_0}{Z + R_0}, \qquad Z = \frac{V}{I}, \qquad I = \frac{2V_{\text{inc}} - V}{R_0}. \tag{2}$$

Every evaluation checks passivity, $|S_{11}| \le 1$.

3.2 Shape adjoint with one factorization per frequency

The design moves mesh node coordinates X, so the gradient needs $\partial K/\partial X$, $\partial M/\partial X$, $\partial S_p/\partial X$ and $\partial \mathbf{b}/\partial X$. geode-fem evaluates its closed-form element kernels in first-order dual arithmetic inside a tensor stack (Tracel AI, 2024): seeding the tangent of one node coordinate returns the exact partial derivatives of every local matrix entry. The UPML weights are held fixed under node

motion. Because $A(\omega)$ is complex symmetric, the adjoint system uses the same operator, and one LU factorization per frequency serves the forward and the adjoint solve. The nodal gradient is $\lambda^{\top}(\partial A/\partial X)\,\mathbf{x}$ plus the explicit dependence of the port drive and of the port-voltage functional on the geometry. When a parameter moves the feed, the port faces move and the port is rebuilt on every moved mesh. A unit test of the moving-feed gradient on a curved mesh, with a source that varies across the port so that the explicit term matters, matches central finite differences to below 10^{-8}; without the explicit term it misses by a relative error of about 3.

3.3 Copper design freedom on a fixed curved board

The board is not a design variable. Each substrate node has chart coordinates (s, y), with $s = R_b\,\mathrm{atan2}(x, z - z_{\mathrm{axis}})$ the mid-plane arc length and y the bend axis. A design applies a chart displacement $(\Delta s, \Delta y)$ and moves the node to angle $\phi_0 + \Delta s/R_b$ and height $y_0 + \Delta y$ at its own radius, so the FR-4 stays a uniform 1.6 mm shell. Every run records the largest change of any substrate node's radius ($\leq 1.4 \times 10^{-14}$ mm). Air nodes follow a harmonic extension of the motion; the UPML and the outer boundary are fixed. The design gradient chains the nodal adjoint gradient through the exact Jacobian of this map.

The 38 parameters, each per unit coefficient, are 8 clamped cubic B-splines per patch edge ($\pm y$, $\pm s$), each moving its edge outward in the surface by 1 mm, ramped in from the feed band and decaying over the 12 mm substrate margin (32); one bump per corner, moving it 1 mm outward along the bisector with support $1 - \mathrm{smoothstep}(d/6\,\mathrm{mm})$ (4); and the probe's inset from the $+y$ edge and its width w_p (2). They are smooth functions of the chart, so the same coefficient vector defines the same copper on every mesh of the family, which lets a design be carried between discretizations. The basis cannot open a hole in the patch. For the scaling study two smaller bases are used: 14 parameters (two linear outline splines per edge, the corners and the feed) and 4 (patch width and length, probe inset and probe width).

3.4 Objective, optimizer and guards

The objective is a soft maximum of $|S_{11}|^2$ over $n_f = 5$ band frequencies,

$$G(\mathbf{c}) = \frac{1}{\beta} \ln \sum_f \exp\!\big(\beta\,|S_{11}(f;\mathbf{c})|^2\big), \qquad \beta = 25, \tag{3}$$

over the coefficient vector $\mathbf{c}$. G exceeds the largest $|S_{11}|^2$ by at most $\ln 5/\beta = 0.064$, and every level reported in this paper is the worst of the band points, not G. L-BFGS (memory 8) with Armijo backtracking and a cap of 1 coefficient unit per step minimizes G. A per-element guard rejects any trial step that would shrink a tetrahedron's signed volume below 0.25 of its base value. Every adjoint descent reported here ends on the guard (one intermediate re-based run of the space-mapping chain ended on a plateau).

Before and after each run the pipeline gradient is compared with a central finite difference along the steepest-descent direction and along random directions. The worst relative error over the runs reported here is 1.5×10^{-5} (tolerance 5×10^{-3}), and a per-group check on the two coarsest meshes passes to 1.9×10^{-7}. The adjoint is implemented for $p = 1$ only, so optimization runs at $p = 1$ and finer models enter through verification.

4 Fixture and Forward-Model Accuracy

Antenna. The initial antenna is a 38×29 mm PEC patch (width along s, resonant length along y) on 1.6 mm FR-4 with a 12 mm substrate and ground margin, fed by a 1.6 mm probe, modelled

Table 1: Where the error sits: the initial patch's $|S_{11}|$ dip under refinement of L1, geode-fem at $p = 1$, with P2/L1 for reference. "Slab" is the element size in the substrate under the patch, "rim" the mean edge length within 0.5 mm of the patch edge, "layers" the median number of elements through the substrate under the patch. Model-dependent values.

Mesh	Slab (mm)	Rim (mm)	Layers	DOFs	Dip (GHz)	vs L1 (MHz)
L1	0.80	0.78	2.00	150,561	2.3118	0
slab 0.533	0.533	0.52	2.54	234,691	2.3273	+15.5
slab 0.40	0.40	0.39	3.44	414,837	2.3404	+28.6
slab 0.30	0.30	0.29	4.19	828,775	2.3480	+36.2
rim 0.40	0.80	0.39	1.99	317,424	2.3420	+30.2
rim 0.20	0.80	0.20	1.99	430,808	2.3569	+45.1
slab 0.30 + rim 0.10	0.30	0.10	4.19	1,738,557	2.3659	+54.1
L1 at $p = 2$ (P2/L1)	0.80	0.78	2.00	804,460	2.3566	+44.8

as a uniform lumped port, inset 7.0 mm from the $+y$ edge (Figure 1). The bend axis is parallel to y, so the curvature runs along the non-radiating width; the sagitta at the patch edge is 1.8 mm. A 40 mm air margin separates the board from the UPML. The target is $|S_{11}| \leq -10$ dB over the 2.400–2.4835 GHz ISM band.

The coarse mesh. The optimization mesh L1 has 150,561 edges, a median of 2.0 element layers through the substrate under the patch and a mean edge length of 0.78 mm at the patch rim. On L1 at $p = 1$ the initial patch's $|S_{11}|$ dip is at 2.3118 GHz; at $p = 2$ on L1 (P2/L1) it is at 2.3566 GHz, 45 MHz higher. A direct LU solve at P2/L1 (804,460 unknowns) takes 132 GiB with geode-fem's direct solver, so refining further at $p = 2$ is not practical with this solver.

The error sits at the rim. Table 1 refines L1 in two ways. Refining the whole slab under the patch to 0.40 mm gives 3.4 layers and, as a side effect, a 0.39 mm rim. Refining only within 0.5 mm of the patch edge to the same rim size, with the 2-layer interior unchanged, lands within 1.6 MHz of it (2.3404 against 2.3420 GHz). The interior layer count does not decide the dip; the rim does, as expected where the field is singular at a conductor edge (Meixner, 1972). A 0.2 mm rim at $p = 1$ (431k unknowns) matches P2/L1 (805k) to 0.3 MHz. Over all seven $p = 1$ rungs the dip follows $f = f_\infty - C\, h_{\mathrm{rim}}^{0.80}$ with a 1.3 MHz rms residual (Figure 2b). The exponent is a fit to the rim edge length on graded meshes, not a proven asymptotic order. The combined rung, which halves the rim to 0.1 mm and refines the slab, moves the dip 9.0 MHz.

An independent solver on the same meshes. Palace (Palace developers, AWS Center for Quantum Computing, 2023) solved the same problem on geode-fem's own meshes, written out with their physical tags, with the same materials, PEC surfaces and uniform 50 Ω lumped port. Its absorber differs (air in the UPML shell, a first-order absorbing outer wall); its driven sweeps use an adaptive reduced-order model with tolerance 10^{-3} on a 1 MHz grid, and GMRES to 10^{-8}. On L1, Palace and geode-fem agree on the dip to 1.1 MHz at $p = 1$ (2.3130 against 2.3118 GHz) and 0.4 MHz at $p = 2$ (2.3570 against 2.3566 GHz); on the strip-fed design of Section 6 the P2/L1 band edges agree to 0.7 MHz (lower) and 0.2 MHz (upper). Palace at $p = 3$ on L1 gives 2.3660 GHz, 8.9 MHz above its $p = 2$ value, and geode-fem's finest rung (slab 0.30 plus rim 0.10 mm) gives 2.3659 GHz (Figure 2a): two different refinements of the same model, order in Palace and rim size in geode-fem, land within the 1 MHz sweep step. Sharing mesh, port model and geometry, this

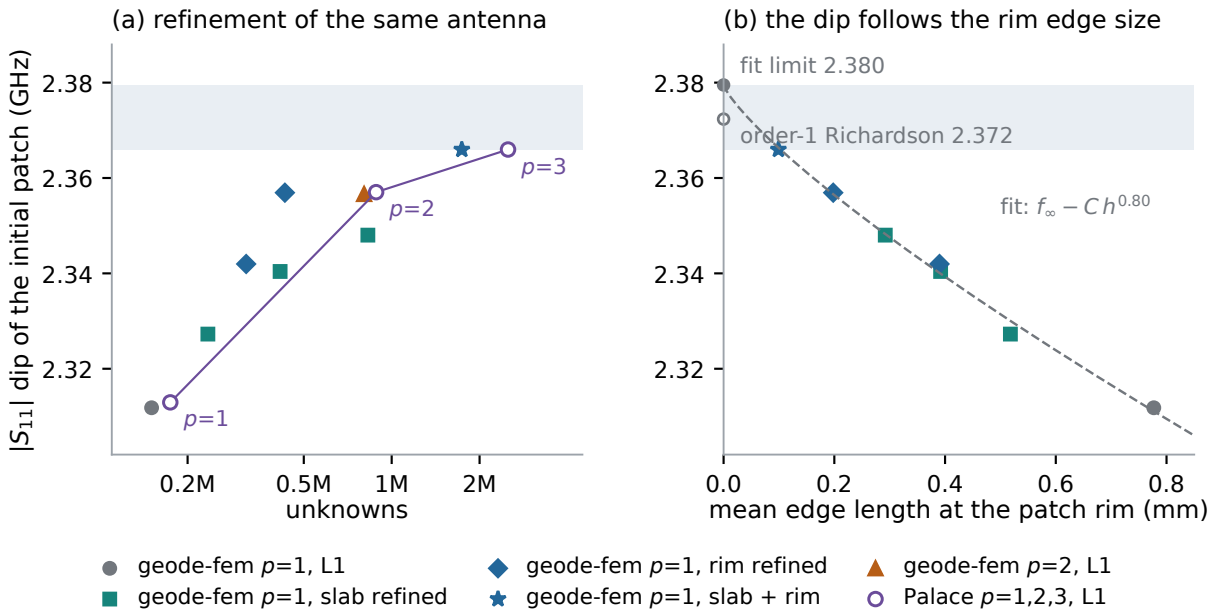


Figure 2: The initial patch's $|S_{11}|$ dip under refinement. (a) Against the number of unknowns: geode-fem at $p = 1$ on L1 and its slab, rim and combined refinements, geode-fem at $p = 2$ on L1, and Palace at $p = 1, 2, 3$ on the same L1 mesh. (b) Against the mean edge length at the patch rim, with the fit $f_\infty - Ch^{0.80}$ (dashed), its limit and the order-1 Richardson step on the 0.39/0.20 mm rim pair. The shaded band, 2.366–2.380 GHz, runs from the finest computed value to the fit's limit; it is an estimate, not a bound. Data: `v6/v6_summary.toml [x1]`.

checks the discretization, not the model.

The mesh-limit resonance. Every refinement computed so far raises the dip. If that continues, the finest computed value, 2.366 GHz, is a lower estimate of the mesh-limit dip; this is an inference from the trend, not a proof. The rim fit puts the limit at 2.380 GHz and the order-1 Richardson step on the rim pair at 2.372 GHz; since the fitted exponent is below 1, the order-1 step is the lower of the two if the fit holds. We therefore estimate the initial patch's mesh-limit dip at 2.366–2.380 GHz, 0 to 14 MHz above the finest computed models.

Port-free resonance. Palace's eigenmode solver, on the same L1 mesh at $p = 2$ with the port gap left open (no feed), finds the patch mode at 2.3408 GHz with $Q = 22.2$ for the initial patch, and at 2.3991 GHz with $Q = 19.3$ for the final design, identified by its substrate energy fraction (0.80–0.83, every other mode at most 0.013). The matched dip on the same discrete model lies above it, by 16.3 MHz for the initial patch and 30.9 MHz for the design, and loading the gap with the 50 Ω port raises the mode by 12.2 and 23.1 MHz and halves its Q (11.5 and 8.6). The feed thus raises the dip above the cavity resonance, as a series-inductive probe would. The Hammerstad transmission-line estimate for the initial patch, 2.433 GHz (Hammerstad, 1975), lies 92.5 MHz above the port-free resonance at $p = 2$ and 76 MHz above the matched dip. The discretization accounts for 9 to 23 MHz of that and the feed moves the dip toward the closed form, so most of the gap is in the cavity model; the finite ground and the closed form's own accuracy are untested candidates.

Absorber, air margin and curvature. Over a ladder of UPML conductivity and thickness, $|S_{11}|$ changes by at most 0.021. Re-meshing the final design with a 60 or 80 mm air margin instead of 40 mm, at $p = 1$ on L1, raises its lower -10 dB edge by 3.0 and 4.3 MHz, monotonically, and its upper edge by 4.2 and 2.9 MHz. The air-margin effect is therefore smaller than the refinement shifts, but of the size of the design's margins (Section 7), and it had not levelled off at 80 mm. The same copper on a flat board (bend radius 10^6 mm, the same mesh node for node) resonates 0.3 MHz from the bent one at $p = 1$, and a complex frequency-shift fit at P2/L1 finds 0.28 MHz. The bend does not detune the patch, but it changes the match: on the strip-fed design the bent board has a 93.3 MHz $p = 1$ -10 dB band against 82.8 MHz flat, and a worst in-band $|S_{11}|$ at P2/L1 of -10.78 against -9.93 dB. We did not investigate the mechanism, and a design optimized flat was not run.

5 The Adjoint Against Gradient-Free Search

Setup. Every optimizer solves one problem: the objective (3) on the unshifted band, the 0.25 volume guard and the L1 fixture at $p = 1$, from $\mathbf{c} = 0$ (-1.47 dB). The finite-difference control runs the adjoint's L-BFGS, line search, step cap and guard with a one-sided difference gradient (step 10^{-5}), one band evaluation per parameter. Separable CMA-ES is geode-fem's own implementation of Ros and Hansen (2008), not cross-checked against a reference implementation on a test function; full-covariance CMA-ES is pycma (Hansen et al., 2026) with its defaults. Both use population $\lambda = 4 + \lfloor 3 \ln n \rfloor$ (14 at $n = 38$) and rank a sample that fails the guard after every feasible sample, by its violation, without a solve. NEWUOA runs in PRIMA with trust radii 0.5 to 10^{-3} and $n + 2$ or $2n + 1$ interpolation points, and a design that fails the guard receives a penalty above any feasible objective, without a solve. For Palace, geode-fem applies the morph and the guard and writes the moved mesh, and Palace solves the five band frequencies at $p = 1$ with GMRES on 16 MPI ranks, a configuration not tuned for speed. The Palace CMA-ES runs reuse the three geode-fem seeds at $\sigma_0 = 0.5$; with the same seed the two solvers make the same evaluations to about evaluation 70, so those six runs are three trajectories, each under two forward models, which isolates the forward solver. One solve is one band frequency with one matrix factorization. The adjoint solve reuses the forward factorization, and the adjoint's start, which yields its first gradient, is counted. The finite-difference control is charged 5 solves per guard-passing trial plus 190 per gradient. The gradient-free runs' separate evaluation at $\mathbf{c} = 0$ is not counted, except as NEWUOA's own first evaluation, and a sample rejected by the guard costs no solve. Each level is the worst of the five band points in that run's own forward model, and a level a run did not reach is censored at the solves it used.

Results with 38 parameters. To reach -9.5 dB the adjoint uses 35 solves (Table 2, Figure 3a). The fastest gradient-free run, a separable CMA-ES seed at $\sigma_0 = 0.5$, uses 385 (11×), NEWUOA with $n + 2$ points 400 (11.4×), and no run fewer. These ratios are against the fastest runs that reached the level; the $\sigma_0 = 1.5$ runs, censored at 80 to 105 solves, bound the -9.5 dB ratio only from below, by 2.3 to 3.0×. To -9 dB the ratio to the fastest run to reach it is 7.7×: full CMA-ES at $\sigma_0 = 0.5$ needs 230 solves against 30, in a single seed, and then 690 to reach -9.5 dB. A larger step does not help: at $\sigma_0 = 1.5$, 89 to 92% of the samples fail the volume guard, a 190-evaluation run makes only 16 to 21 feasible evaluations, and the seed given 1000 evaluations (1,865 solves) ends at -8.81 dB, so the guard bounds the useful step. The smaller NEWUOA interpolation set shortens the start-up but not the time to -9.5 dB (400 against 420 solves); that run also reaches -10.06 dB, the best level of any 38-parameter run. The $\sigma_0 = 0.5$ CMA-ES and NEWUOA runs had 4 to 8 times the adjoint's solves, so the budgets favour them.

At -10 dB the comparison is weak. Two gradient-free runs reach it, at 650 and 695 solves

Table 2: Solves to fixed worst-of-band levels with 38 parameters on the L1 optimization model ($p = 1$), from $\mathbf{c} = 0$. Slashes separate seeds; the two $\sigma_0 = 0.5$ separable CMA-ES rows use the same three seeds in the same order and coincide to about evaluation 70. "Used" is the solves a run used, where a censored entry (–) stops. Budgets: 190 evaluations; 114 for the first $\sigma_0 = 0.5$ seed on geode-fem, 1000 for the long run, 2,500 solves for finite differences.

Optimizer, setting	Forward	Used	Solves to			Best
			−9 dB	−9.5 dB	−10 dB	(dB)
L-BFGS, adjoint	geode-fem	125	30	35	60	−10.04
L-BFGS, finite differences	geode-fem	2,535	1,170	1,365	2,340	−10.04
sep-CMA-ES, $\sigma_0 = 0.5$	geode-fem	535/785/690	405/385/395	520/385/395	–/695/–	−9.56/−10.02/−9.77
same seeds	Palace	860/780/660	405/385/395	805/385/570	–/–/–	−9.58/−9.89/−9.90
sep-CMA-ES, $\sigma_0 = 1.0$	geode-fem	460	380	–	–	−9.29
sep-CMA-ES, $\sigma_0 = 1.5$	geode-fem	95/100/105	–	–	–	−6.29/−1.48/−1.30
1000 evaluations	geode-fem	1,865	–	–	–	−8.81
CMA-ES, $\sigma_0 = 0.5$	geode-fem	695	230	690	–	−9.82
CMA-ES, $\sigma_0 = 1.5$	geode-fem	80	–	–	–	−3.51
NEWUOA, $n + 2$ points	geode-fem	945	380	400	650	−10.06
NEWUOA, $2n + 1$ points	Palace	945	415	420	–	−9.95

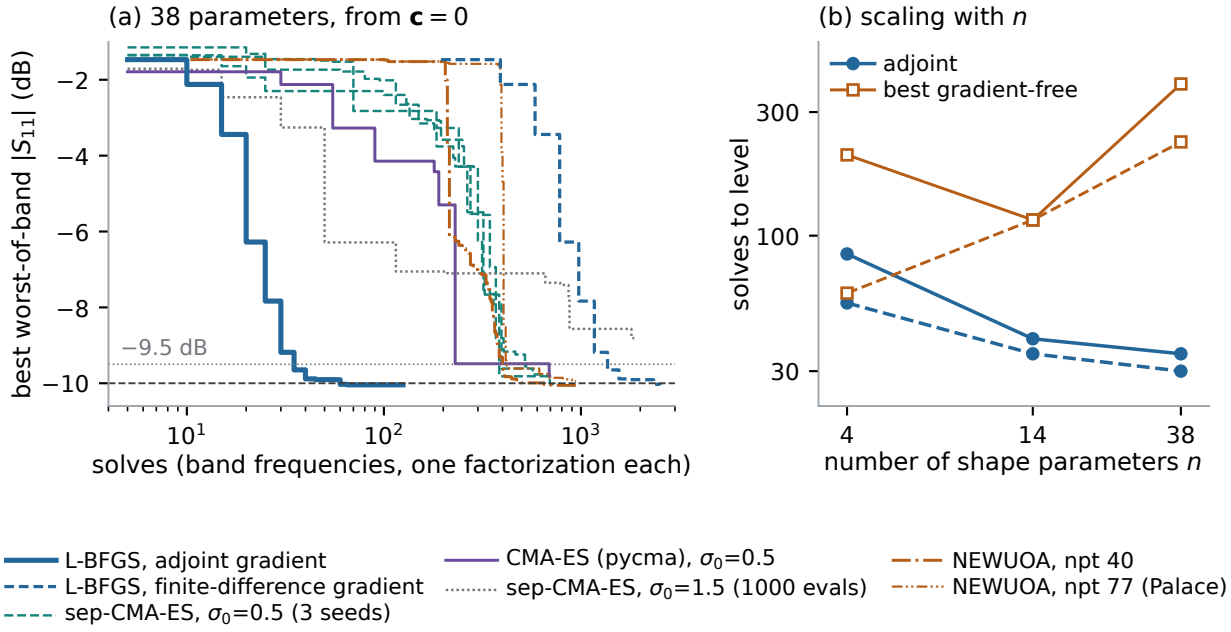


Figure 3: (a) Best worst-of-band $|S_{11}|$ so far on the L1 optimization model against solves, with 38 parameters, for the adjoint and finite-difference L-BFGS and the gradient-free runs of Table 2 (the Palace CMA-ES runs, which share seeds with the dashed geode-fem runs, and the short $\sigma_0 = 1.5$ runs are omitted). (b) Solves to −9.5 dB (solid) and −9 dB (dashed) against the number of parameters, for the adjoint and for the best of the 5 to 11 gradient-free runs at each n. Data: `tuned/`, `designc/c4/`, `v6/x4/`, `v6/palace_loop/`.

against the adjoint's 60, and the rest are censored. The forward models agree to 0.004 dB on the 41 designs both evaluated (the first CMA-ES generation of each seed, 42 samples with one rejected by the guard in both) but differ by up to about 0.25 dB at optimized designs (Palace reads the adjoint's final design at −9.80 dB against geode-fem's −10.04 dB), the size of the gap between reaching and missing −10 dB, so we derive no ratio from the −10 dB column.

The finite-difference control. With a forward-difference gradient, L-BFGS takes the adjoint's path. The two gradients agree to 4.6×10^{-7} (relative) at the start and 5.2×10^{-4} at the final design, the worst-of-band value at each of the 12 steps matches the adjoint run's to 1.7×10^{-5} dB, and both reach -10.04 dB at step 12. A step costs 195 solves against the adjoint's 5, so the ratio is $39\times$ at every level, and the 1,365 solves to -9.5 dB are 2.6 to 3.55 times the 385–520 of separable CMA-ES at $\sigma_0 = 0.5$ and more than every gradient-free run that reached the level. With the optimizer, basis, guard and iterates fixed, only the cost of the gradient changes, so this ablation credits the solve savings to the adjoint; a finite-difference gradient on 38 parameters is worth less than none. The design itself comes from the optimizer and the parametrization, which give the same iterates with either gradient.

Scaling with the number of parameters. From the same cold start with 4 and 14 parameters, against the best of three separable CMA-ES seeds at $\sigma_0 = 1.5$ and NEWUOA with $n + 2$ and $2n + 1$ points, the ratio to the fastest run to reach each level is $1.1\times$ at -9 dB and $2.4\times$ at -9.5 dB with 4 parameters, $3.3\times$ and $2.9\times$ with 14, and $7.7\times$ and $11\times$ with 38 (Figure 3b). With 4 the adjoint stops on the guard at -9.85 dB and three gradient-free runs end lower (-9.92 to -9.97 dB); with 14 it reaches -10.00 dB in 60 solves and the best gradient-free run ends at -9.99 dB. At 4 and 14 parameters separable CMA-ES ran at $\sigma_0 = 1.5$ only, the worst setting at 38, which may favour the adjoint on the CMA-ES side, but NEWUOA, which has no σ_0, gives the best gradient-free count at both. Given the 38-parameter chain's final band shift and feed re-base in advance (Section 6), the four-parameter rectangle does reach -10.33 dB at P2/L1, against -10.78 dB for the 38-parameter design.

The forward solver and the wall-clock. The paired Palace runs reach -9 dB at the same counts as their geode-fem twins, so the result does not depend on geode-fem's forward model, though Palace's band evaluation is the cheaper one (64.4 against 72.1 s on an r6i.4xlarge). To -9.5 dB the Palace-driven runs took a measured 84–172 min (CMA-ES) and 90 min (NEWUOA) against an estimated 11.1 min for the adjoint (solves times measured per-solve cost). These are serial times on one host; CMA-ES would spread across hosts, while L-BFGS spreads only over the five band frequencies, so the solve count is the hardware-independent measure.

6 Design Procedure

Frequency space mapping. A design optimized on L1 resonates higher on a finer model. We model the fine response as the coarse response shifted in frequency, $S_{\text{fine}}(f) \approx S_{\text{coarse}}(f - \delta)$, fit δ by least squares over the fine model's points, and re-optimize on the coarse model for the band shifted down by δ. For the cold L1 optimum of Section 5 the fitted L1-to-P2/L1 shift is 47.85 MHz (complex misfit 0.23–0.38 unshifted, 0.018–0.025 shifted); without it, the cold L1 optimum (-10.04 dB on L1) gives -4.76 dB at P2/L1.

Feed re-basing. Moving the feed toward the $+y$ edge compresses the mesh between them, and the cold runs stop there at the volume guard. We then regenerate the mesh with the port at its moved position and carry the design over by shifting the two feed coefficients. After re-basing, the optimizer widens the patch until the guard binds in the air 5.5 mm outside a $\pm s$ edge, which is a non-radiating edge.

Three designs. Table 3 lists the designs. Design A ends the space-mapping chain centred for P2/L1 with the feed width free, which widened the port into a 4.27 mm strip. At P2/L1 it reaches

Table 3: The designs. δ is the band shift of the final descent. "Solves" counts the driven solves of that descent as recorded, including its gradient checks, a different count from Table 2's. "Fine model" is the model the design was centred for.

Design	Feed	δ (MHz)	Steps	Solves	Fine model
A	4.27 mm strip (free width)	44.94	9	125	P2/L1
B	1.6 mm probe (pinned)	44.94	10	130	P2/L1
C (final)	1.6 mm probe (pinned)	57.0	10	125	refined rim, Palace $p = 3$

Table 4: Design C on each model, in order of refinement, with design A on Palace $p = 3$ for comparison. "Worst" is over the nine in-band points (2.400 to 2.4835 GHz in 10.4375 MHz steps). Band crossings are interpolated linearly in dB. Margins are 2.400 GHz minus the lower crossing and the upper crossing minus 2.4835 GHz. Design C's meshes are the re-based fixtures with the port at its moved position, so their counts differ slightly from Table 1's. Model-dependent values.

Model	DOFs	Worst (dB)	-10 dB band (GHz)	Coverage	Margins (MHz)
geode-fem $p = 1$, L1	150,541	-4.94	2.3393–2.4304	36%	60.7 / -53.1
geode-fem $p = 2$, L1	804,336	-9.08	not swept	not swept	not swept
Palace $p = 2$, L1	884,536	-9.08	2.3847–2.4770	92.2%	15.3 / -6.5
geode-fem $p = 1$, rim 0.2	431,528	-9.23	2.3849–2.4782	93.6%	15.1 / -5.3
geode-fem $p = 1$, rim 0.1 + slab	1,738,279	-10.73	2.3934–2.4881	100%	6.6 / 4.6
Palace $p = 3$, L1	2,503,077	-10.46	2.3935–2.4864	100%	6.5 / 2.9
Palace $p = 2$, rim 0.2	2,465,984	-10.32	2.3981–2.4912	100%	1.9 / 7.7
Design A, Palace $p = 3$	2,729,331	-9.56	2.4029–2.4975	96.5%	-2.9 / 14.0

-10.78 dB with a 94.9 MHz band, 2.3938–2.4887 GHz. At Palace $p = 3$ its band moves up to 2.4029–2.4975 GHz, the lower edge misses 2.400 GHz by 2.9 MHz, and the worst in-band value is -9.56 dB, so a design centred on P2/L1 misses once the rim is resolved. Design B re-runs A's last step with the probe width pinned at 1.6 mm and reaches -10.62 dB at P2/L1 (-10.65 dB with a 1.27 mm probe), so the wide strip was not what carried the bandwidth.

Design C re-centres B for the refined model with a band shift of 57.0 MHz. The shift adds three terms: 44.94 MHz from L1 to P2/L1, 8.7 MHz for design A's $p = 2$ to $p = 3$ dip shift in Palace, and 3.4 MHz further down because every refinement so far has moved the band up. A parallel run with the first two terms only, 53.6 MHz, was set aside by a prediction from the fine models. For each run we added the initial patch's step from the 0.2 mm-rim mesh to Palace $p = 3$, about 9 MHz, to its lower edge on that mesh; this predicts a lower-edge margin at $p = 3$ of about 2.6 MHz for the 53.6 MHz run and about 6 MHz for C. Because Palace $p = 3$ informed both the shift and this choice, the geode-fem mesh that was not used in it carries the independent check in Section 7. The descent to C ran 10 steps and 125 solves, from -8.73 to -10.64 dB on its shifted L1 band, and stopped on the guard with the gradient norm down from 0.143 to 2.2×10^{-3}; the gradient matched finite differences to 1.2×10^{-6}. As built, C is 29.00 mm long and 49.72 mm wide on the centre line, with a barrel outline whose $\pm s$ edges bow out most at mid-length, and its 1.6 mm probe sits 5.14 mm from the $+y$ edge.

7 Results

On the finest models. On Palace at $p = 3$ on L1, design C has a worst in-band $|S_{11}|$ of -10.46 dB, at 2.4835 GHz, and a contiguous -10 dB band from 2.3935 to 2.4864 GHz, 93.0 MHz wide, which covers the 83.5 MHz ISM band with 6.5 MHz to spare at 2.400 GHz and 2.9 MHz at 2.4835 GHz

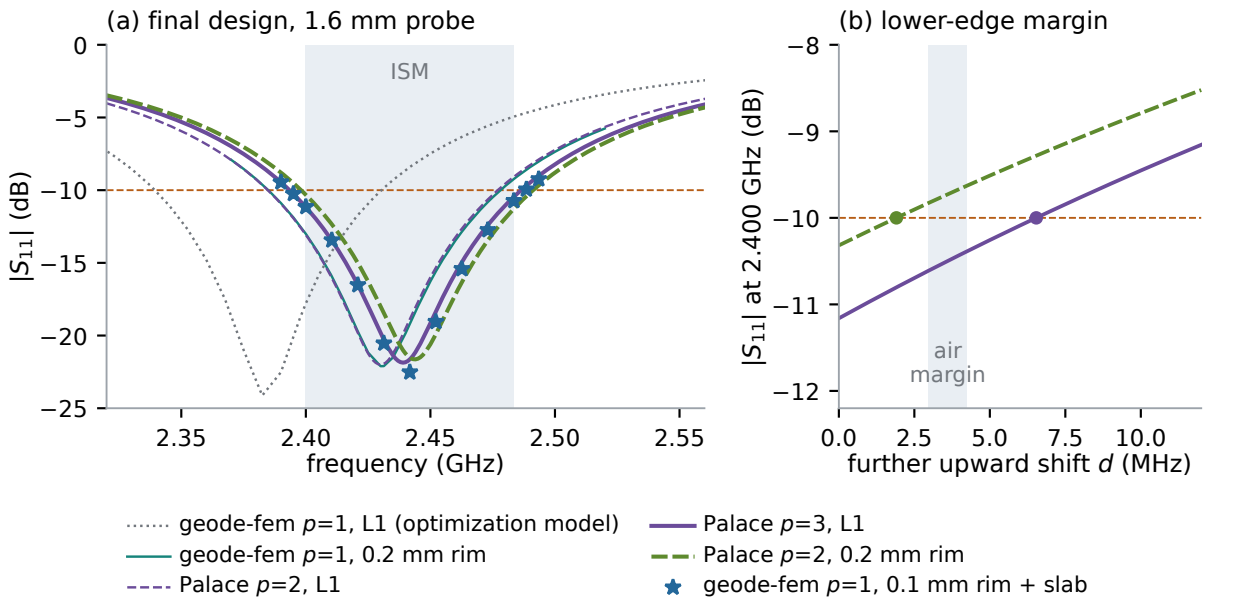


Figure 4: (a) Return loss of design C. Palace at $p = 3$ and $p = 2$ on the design's L1 mesh and at $p = 2$ on its 0.2 mm-rim mesh (1 MHz grids); geode-fem at $p = 1$ on the combined 0.1 mm-rim and 0.3 mm-slab mesh (13 points) and on the 0.2 mm-rim mesh; and geode-fem at $p = 1$ on L1, the optimization model, where the design sits 57 MHz low by construction. (b) $|S_{11}|$ at 2.400 GHz if a whole Palace response moves up by a further d, read from the computed response at $2.400 - d$ GHz; the dots mark the lower-edge margins, 6.5 and 1.9 MHz. The shaded range is the rise of the lower edge with a 60 or 80 mm air margin (geode-fem, $p = 1$, L1). Data: `recentre/`, `designc/c3/`, `tuned/tuned_summary.toml`.

(Table 4, Figure 4a). geode-fem's finest mesh, the combined 0.1 mm rim and 0.3 mm slab (1.74M unknowns), gives −10.73 dB and a 94.7 MHz band, 2.3934–2.4881 GHz. The two lower edges agree to within the sweep resolution (the interpolated values differ by 0.02 MHz); the upper edges differ by 1.7 MHz, partly because geode-fem's upper crossing is interpolated across a 5 MHz gap. Palace at $p = 2$ on the 0.2 mm-rim mesh (2.47M unknowns), the most refined model computed for the design, gives −10.32 dB and a band of 2.3981–2.4912 GHz: its lower edge sits 4.6 MHz above Palace $p = 3$ on L1, which leaves 1.9 MHz of lower-edge margin. The two models are different refinements of comparable size (one more order on L1, and the rim at $p = 2$), so the 4.6 MHz shows that $p = 3$ on L1 does not yet resolve the rim. From $p = 2$ to $p = 3$ on L1 the design's dip rises 9.1 MHz and its lower band edge 8.7 MHz; the initial patch's dip rises 8.9 MHz over the same step, so the design shifts like the patch.

On the refined meshes the air is moved on cylinders (the default morph inverts elements on the combined mesh); on the 0.2 mm-rim mesh the two air motions agree to 0.09 MHz at the lower edge.

The lower-edge budget. Table 5 collects what moves the lower edge. The two measured terms, added, take more than the margin: $6.5 - 4.6 - 4.3 = -2.4$ MHz, a lower edge near 2.402 GHz, and on the 0.2 mm-rim response moved up by 4.3 MHz, 2.400 GHz sits at −9.63 dB (Figure 4b). The sum assumes that the terms add, and the air-margin term was measured on the coarse model. The rim step (4.6 MHz) is below both drift estimates of the initial patch above Palace $p = 3$ (order-1 Richardson 6.4 MHz, rim fit 13.5 MHz), so it need not be the last of them; Palace $p = 3$ on the 0.2 mm-rim mesh was not run. So design C covers 2.400–2.4835 GHz at −10 dB on the three finest discretizations computed, but once refined further and given a larger air margin we estimate that it

Table 5: What moves design C's lower −10 dB edge, against its margin on Palace $p = 3$, L1. A negative entry uses up margin.

Term	MHz	Tier, source
Lower-edge margin, Palace $p = 3$, L1	+6.5	model-dependent, Table 4
Rim refinement at $p = 2$ (Palace, 0.2 mm rim)	−4.6	model-dependent, Table 4
Air margin, 40 → 60 / 80 mm	−3.0 / −4.3	model-dependent at $p = 1$ on L1; transfer estimated
Further refinement above the 0.2 mm-rim model	not computed	initial patch: 6.4–13.5 above Palace $p = 3$, estimated
Coaxial feed in place of the lumped port	not quantified	
Copper loss in place of PEC	not quantified	

misses the lower band edge by a few megahertz, and we do not claim that it meets the ISM band.

Where the descent stopped. The descents to A, B and C stop on the volume guard. For C the gradient norm fell 65-fold first, so C may be near a stationary point of the guarded problem (for the rectangle it grew, 0.023 to 0.074), and the widened patch comes within about 6 mm of the 62 mm board edge. Remeshing at the widened outline would relieve the guard; it was not run.

Radiation efficiency and gain. For design C we computed the accepted port power, the Poynting flux through a closed surface in the air box, the FR-4 loss and the broadside gain by a near-to-far-field transform at $p = 1$, on L1 at the band it was optimized on and on the 0.2 mm-rim mesh across the ISM band. The power balance closes to 0.8% on L1 and 1.6% on the rim mesh. The radiation efficiency is 0.61–0.63, the dielectric loss fraction 0.37, and the broadside gain 4.45–4.66 dBi with a front-to-back ratio of 13–15 dB. With the worst in-band match on Palace $p = 3$ (a mismatch loss of 0.41 dB) the realized gain is at least 4.13 dBi. With PEC conductors this efficiency counts radiation against dielectric loss only; copper loss would lower it and the gain.

Bandwidth against a closed form. On the voltage-standing-wave-ratio-2 criterion, Palace gives design C a bandwidth of 99.0 MHz at $p = 3$ (design A: 100.3 MHz at $p = 2$ and 101.0 MHz at $p = 3$). The closed-form formulas of Jackson and Alexopoulos (1991) give 70.4 MHz for a 49.75 × 28.80 mm rectangle on this substrate, and 67.9 MHz at the area-equivalent width of 45.4 mm. Since $\tan\delta$ is the only loss, design A's dielectric loss fraction gives the model inequality $Q_t \geq 0.362/\tan\delta = 18.1$ (not a bound in the sense of Section 1) against the closed form's $Q_t = 24.5$ and radiation Q of 48.0. We read this, as an interpretation, as the model patch radiating more strongly than the infinite-ground closed form, perhaps through its finite 62 × 53 mm ground, which we did not test.

Cost. The final descent took 26 min on a 32-vCPU host, the finite-difference control 4.7 h on a 16-vCPU host (1,164 to 1,211 s per gradient on 8 threads), the Palace $p = 3$ sweep of design C 3.5 h on 16 MPI ranks and the 0.2 mm-rim sweep 1.5 h. geode-fem's direct solver needed about 8 min per frequency at 204 GiB on the combined mesh.

8 Limitations and Future Work

Feed, losses, refinement and scope. The probe is a uniform lumped port of the pin's width across the substrate, not a coaxial probe. A coaxial feed adds the pin's series inductance and a coax-to-substrate transition. Loading the gap with the lumped port alone moves design C's mode by

23 MHz, so a coax-fed model could move the match by megahertz, as much as the margins; Palace supports wave ports, and a coax-fed model was not run. Conductors are PEC, and the design has not been fabricated or measured. The drift above the 0.2 mm-rim $p = 2$ model is estimated, not computed. A $p = 2$ shape adjoint, or optimizing on a rim-refined mesh, would remove the coarse-to-fine shift that space mapping corrects here, and re-centring with the air-margin term included is the next step. The optimizer comparison is one problem from one start on the coarse model, with one full-CMA-ES seed per step size and one NEWUOA trust radius; a surrogate-assisted or Bayesian optimizer may narrow the gap, and whether the adjoint helps at the refined level is not tested.

Comparison with FDTD. A matched-geometry Meep (Oskooi et al., 2010) model of the initial antenna places the bent patch's resonance near 2.09 GHz, about 250 MHz below the finite-element models. The difference comes from that model's voxelization of its one-cell metal shells without subpixel averaging: on the curved board the normal and tangential electric field see gaps of different heights, which lowers the resonance. With the metal aligned to the grid at 0.5 mm cells, Meep's flat board resonates at 2.335 GHz against geode-fem's 2.343 GHz on a comparable flat model, and the bend moves the Meep dip by −2.6 MHz against geode-fem's −1.0 to −1.6 MHz. A model of the gap mismatch reproduces the artifact to about 1–2%, too coarse to resolve megahertz bend shifts. [**PENDING: results of the corrected Meep harness on the bent initial patch, at its grid resolutions, against geode-fem and Palace.**] Until these are in, we make no comparison of cost or accuracy with FDTD, including a conformal FDTD scheme (Dey and Mittra, 1997; Zagorodnov et al., 2003).

9 How geode-fem Was Designed and Built

geode-fem implements the whiteroom operator specification of Crutcher Dunnavant (Dunnavant, 2026), whose four layers restate Palace's cited source as pure functions, fused primitives, tensor operations and a small graph-evaluation calculus. Dunnavant also gave design input on the solver's operator layer and contributed code: the migration to Rust 2024, the Metal (Apple GPU) backend, the present layout of the core crate and the examples, and the geode-util crate. The rest of the code, the experiments and the text of this paper were produced by AI models working as role-separated agents: Anthropic's Claude (Opus, Fable and Sonnet models, through Claude Code) and OpenAI's GPT Sol and GPT Astra models (via Codex). The agents were coordinated by Loom (Walters, 2025), an issue, worktree and pull-request workflow, and by Squad, a message room the agents share. Nearly every change was reviewed by a separate agent session. The agent reviews of the experiments behind this paper recomputed the reported numbers from the archived records, and the script that checks this paper's text against those records was mutation-tested. The human author directed the research, made every scoping and claim decision, and takes responsibility for the content.

10 Reproducibility

Every result in this paper is read from an archived, scripted artifact.[1] Table 6 lists them, relative to `benchmarks/patch_antenna_conformal/`. Each run record holds the exact command and the SHA-256 of its mesh; the meshes are regenerated from YAML recipes in `reference/gmsh/` with gmsh 4.15.2, and runs on re-based fixtures need the `--feed-support-legacy` flag. Many records carry no generating commit (they ran outside a git checkout); the code is merged at geode-fem commit `99646f06`, and the tuned baselines and air-margin runs used `dcd8df60`. Palace is commit

[1] The geode-fem source is public at `https://github.com/rjwalters/geode-fem`.

Table 6: Artifacts behind each result, relative to `benchmarks/patch_antenna_conformal/`.

Result	Artifact
Fixture, refinement, Palace cross-check, curvature (Section 4)	`baseline_2g4_L1.toml`, `convergence.toml`, `upml_ladder_L1.toml`; `v6/v6_summary.toml`, `v6/x1/`, `v6/x2/`
Baselines and scaling (Section 5)	`tuned/tuned_summary.toml`, `tuned/`; `v6/palace_loop/palace_loop_summary.toml`, `v6/x4/`; `copper/L1_copper_38.toml`; `designc/c4/`
Designs A to C (Sections 6, 7)	`copper/g2_summary.toml`, `v6/x3/`, `v6/x5/`; `recentre/recentre_summary.toml`, `recentre/palace/`; `designc/designc_summary.toml`, `designc/c1/`–`c3/`; `tuned/b5/`
FDTD comparison (Section 8)	`reference/meep/docker/DIAGNOSIS.md`; `benchmarks/fdtd_density_baseline/meep_runs/diag/diagnosis.json`
Generator and tests	`crates/geode-core/examples/patch_conformal_nested_dof.rs`; `crates/geode-core/tests/patch_conformal_*.rs`

`fba6a5b`, built from `reference/palace/docker/Dockerfile`; PRIMA was built from libprima commit `43863c69` (package `0.7.3.dev1801+g43863c69c`), and pycma is 4.5.0. geode-fem ran on AWS Graviton4 hosts (r8g.4xlarge and r8g.8xlarge), Palace and the cost measurements on r6i.4xlarge. The figure scripts in `figures/src/` read the artifacts, `provenance.md` maps every number to its source field, and `verify_evidence.py` checks the text against them.

11 Conclusion

On the copper design of a bent 2.4 GHz patch, a finite-element shape adjoint reaches a worst-of-band $|S_{11}|$ of -9.5 dB on the coarse optimization model with 11 times fewer solves than the fastest tuned gradient-free run to reach that level, and -9 dB with 7.7 times fewer than the fastest run to reach -9 dB. The ratio is near one with 4 parameters and grows with their number. A finite-difference gradient on the same path costs more than gradient-free search, so the savings are the adjoint's, and the optimum is no better: gradient-free search reaches the same level given enough solves. The design, re-centred on the rim-resolved model, with a 1.6 mm probe on a standard board, covers the ISM band at -10 dB on the three finest models computed, but its lower-edge margin falls to 1.9 MHz on the most refined one and a larger air margin takes another 3 to 4 MHz, so we estimate that it misses 2.400 GHz by a few megahertz.

Acknowledgments. We thank Crutcher Dunnavant for the whiteroom specification, for his design input to the solver's operator layer and for his contributions to the codebase. The AI-orchestrated development of geode-fem and of this paper is described in Section 9.

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